

**FIRST COURSE HANDOUT,
ELEMENTARY STOCHASTIC PROCESSES II (MTH313M),
2025-26 EVEN SEMESTER (SECOND HALF)
3 - 1 - 0 - 0 [6 CREDITS]**

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1. PRE-REQUISITES AND COURSE OBJECTIVES

Pre-requisites: MSO201 Probability and Statistics or MSO205 Introduction to Probability or equivalent

Objectives: This course aims to cover the basics of continuous time Stochastic processes, specifically Poisson processes and general jump Markov processes. This is a DC course for M.Sc (Statistics) students.

2. COURSE CONTENTS AND REFERENCES

2.1. Course contents.

- (Topic 1) Continuous time Stochastic Processes (focus on Counting process), Kolmogorov Consistency/Existence Theorem (statement only)
- (Topic 2) Poisson Process (PP)
- Definition of a PP as a counting process
 - Alternative definition of a PP
 - Inter-arrival and Waiting times for a PP, Coupon Collector's problem
 - Order Statistics and PP
 - Non-homogeneous PP, properties involving the arrival times
 - Compound PP
- (Topic 3) Continuous time Markov Chains (Discrete State space) or Jump Markov Processes
- Definition (focus on homogeneous Markov Chains)
 - Examples: Birth and Death processes, Poisson Process as a Pure Birth process, M/M/c queues, Linear growth Models with immigration, Yule Process
 - Transition Probability function, Chapman-Kolmogorov equations (forward and backward)
 - Limiting probabilities (may be an overview only, heuristic proof may be given with some motivation using the first module)
- (Topic 4) Brownian Motion (BM)
- Definition of a BM, BM as a Markov process

- Gaussian processes
- Properties of a BM
 - Covariance function, Invariance properties
 - Path properties - using the Kolmogorov Continuity Theorem (statement only)

(Topic 5) Special topics (one of the following two topics to be covered) lectures]

- Topic 1: Processes related to Brownian motion
 - Brownian bridge, Application of Brownian bridge in the study of empirical processes, Donsker's Invariance Principle (statement only)
 - Stopping times, Hitting times, Strong Markov Property, Arc-Sine laws
 - Variants of BM: involving absorption and reflection, Geometric BM, Integrated BM, involving drift
- Topic 2: Branching processes
 - A little history of branching processes. The Galton-Watson Branching Processes (GWP). Probability generating function of GWP. Moment generating functions and moment calculations
 - Sub-critical, critical and super-critical scenario and the probability of extinction. Examples. Introduction to Continuous time Markov Branching Processes.
 - Examples of Continuous time Markov Branching Processes – revisit to Yule Process or Binary Fission, Birth & Death Process. Generating functions. Sub-critical, critical and super-critical scenarios.

2.2. Reference materials. References:

- Samuel Karlin and Howard M Taylor: A First Course in Stochastic Processes, 2ed, Academic Press, 1975.
- Sheldon M Ross: Stochastic Processes, John Wiley and Sons, 1996.
- Achim Klenke: Probability Theory, Springer-Verlag, 2008.

Supplementary texts:

- Robert B. Ash: Basic Probability Theory, John Wiley and Sons, 1970.
- Etienne Pardoux: Markov Processes and Applications, John Wiley and Sons, 2008.

2.3. Important dates.

Week #	Dates	Remarks
1	March 9 - 13	
2	March 16 - 20	Drop deadline: March 20, Holiday: March 21
3	March 23 - 27	Holiday: March 26
4	March 30 - April 3	Holidays: March 31 & April 3
6	April 6 - 10	Working Saturday: Friday Schedule
7	April 13 - 17	
8	April 20 - 24	End of the course
End Sem Exam	April 27 - May 7	As per schedule announced by DOAA
Make-up Exam	May 10 - 11	As per schedule announced by DOAA

3. LECTURE SCHEDULE AND OFFICE HOURS

- Lectures: MW 15:30-16:45 hrs (venue: L11).

- Tutorials: F 12:00-13:00 hrs (venue: L11).
- Office Hours: Students should communicate their need to discuss doubts by email and take appointments for the same.

4. WEIGHTAGES FOR DIFFERENT COMPONENTS OF EVALUATION (OUT OF 100)

Component Name	Weightage
Take-home Assignments	2x10 = 20
End-semester Examination	60
Quiz	20

- End Semester Examination: The examination shall be conducted as per the schedule announced by DOAA. The syllabus, question pattern and other details shall be announced later on.
- Quiz: There shall be one quiz. Syllabi and other details of the quizzes shall be announced from time to time.
- Take-home Assignments: There shall be two take-home assignments, with designated due dates. One question from the question set in each of the assignments shall be chosen by the instructor for grading.
- Exercises/Practice Problems: Exercises and/or practice problems shall be provided on a weekly basis and students are expected to solve them on their own. There is no need to submit the solutions – some of the problems shall be discussed during the weekly tutorial sessions. However, students are strongly encouraged to discuss amongst themselves and also solve problems from the books mentioned above.

5. COURSE POLICIES: ATTENDANCE, GRADING, WITHDRAWAL

- There is no extra weightage for attendance. However, students are strongly advised to attend all the lectures.
- The policy of relative grading will be followed for awarding final grades. However, if the highest marks is below 92, then A* grade may not be awarded.
- No make-up opportunity shall be provided for a missed take-home assignment or a quiz.
- There will be no proration option for a missed take-home assignment.
- In the case where a student misses a quiz due to a bonafide non-emergency situation, then a request for proration may be considered. However, such a request must be made well in advance and the final decision to allow prorating rests with the instructor.
- For missing a quiz due to an emergent situation, request for proration must be made as soon as possible after the quiz date. In case of a medical emergency, the student must present a letter from the doctor stating that the student was not in a condition to take the quiz. The final decision to allow prorating rests with the instructor.
- Make-up option for the End-semester examination will be as approved by DOAA. Applications for the same need to be supported with proof, such as, a medical certificate or a proof of sanctioned leave.
- Any dishonest practice during examinations or quizzes will be reported to DOAA and appropriate action would be taken to penalize such action.
- Students are allowed to withdraw from the course as per the guidelines set by DOAA.